FRONT DRIVE AXLE/DIFFERENTIAL

DRIVE PINION SEAL (G1557050)

REMOVAL AND INSTALLATION

51.20.02	OIL SEAL - PINION - DIFFERENTIAL HOUSING - RENEW	3000 CC, TDV6, DIESEL, WITH PARTICULATE FILTER, WITH DIESEL EXHAUST FLUID	4.8	USED WITHINS	+

SPECIAL TOOL(S)

100-012

Slide Hammer

205-067

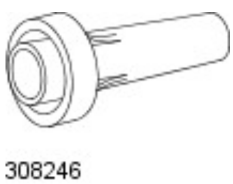
Preload Gauge

303-1300

Remover, Crankshaft Seal

307-520

Installer, Output Shaft Seal



308-246

Installer, Front Seal

JLR-205-1010

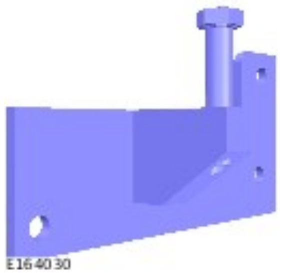
Remover, Bearing

JLR-205-1013

Installer, Bearing

JLR-205-890

Support, Differential



JLR-205-890-2

Pinion Support, Differential Case

E145957

JLR-205-995

Holder, Front Differential Pinion

E145958

JLR-205-996

Remover/Installer, Differential Pinion Nut

E145963

JLR-205-997

Installer, Front Differential Seal

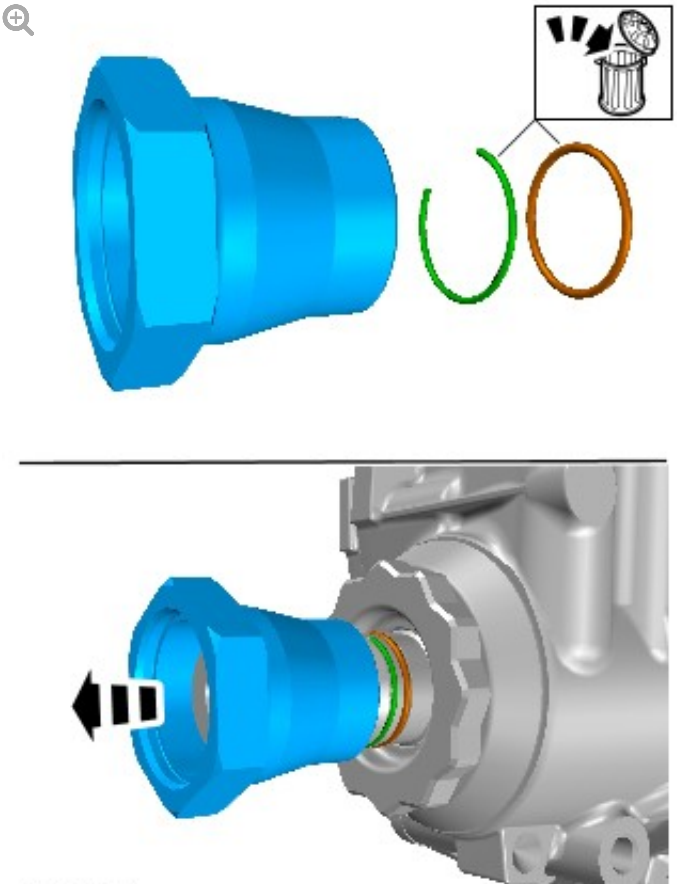
REMOVAL

CAUTION:

Torque to rotate should always be measured and recorded at step 9 before removing the drive pinion nut.

1. Refer to: Differential Draining and Filling (205-03 Front Drive Axle/Differential, General Procedures).
2. Refer to: Axle Assembly (205-03 Front Drive Axle/Differential, Removal and Installation).
3. **NOTES:**

   - Remove and discard the drive pinion circlip.
   - Remove and discard the drive pinion o-ring seals.



E162513

Remove the drive shaft nut.

4.

E162336

5.

CAUTION:

Be prepared to collect escaping oil.

E162337

6.

WARNING:

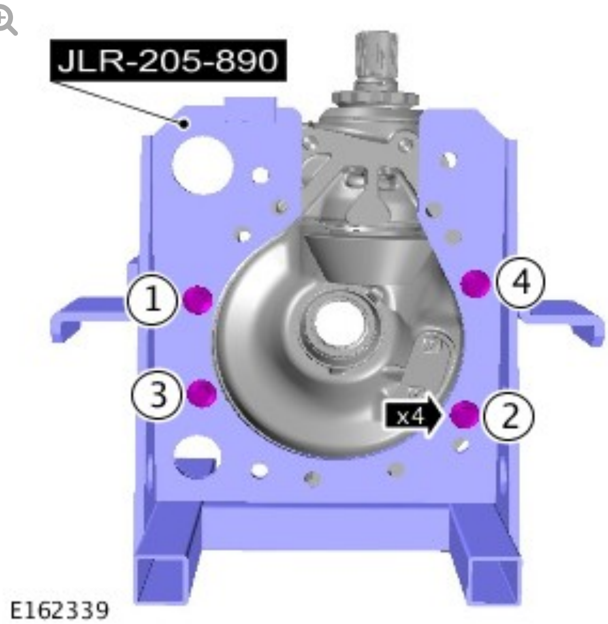
Remove sealant from casing threads and bolts.

NOTE:

Clean all the mating faces thoroughly and check for damage.

7. **NOTE:**

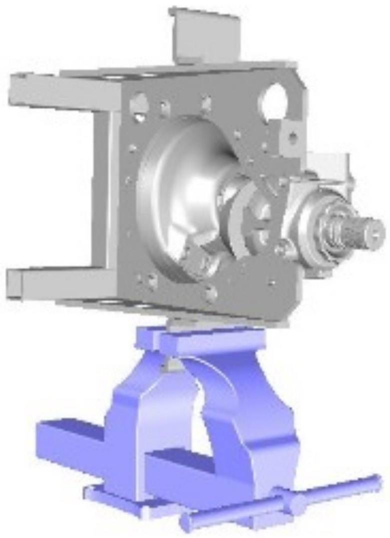
Tighten the retaining bolts in the sequence shown.



Special Tool(s): JLR-205-890

Torque: **45 Nm**

8.

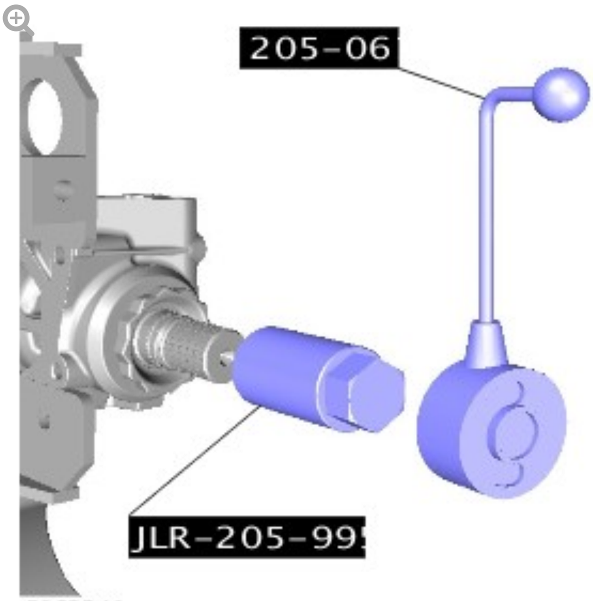


E162340

9.

NOTES:

- Check and record torque to rotate before removing the drive pinion assembly any further.
- The special tool 205-067 should be rotated clockwise to check the torque to rotate.

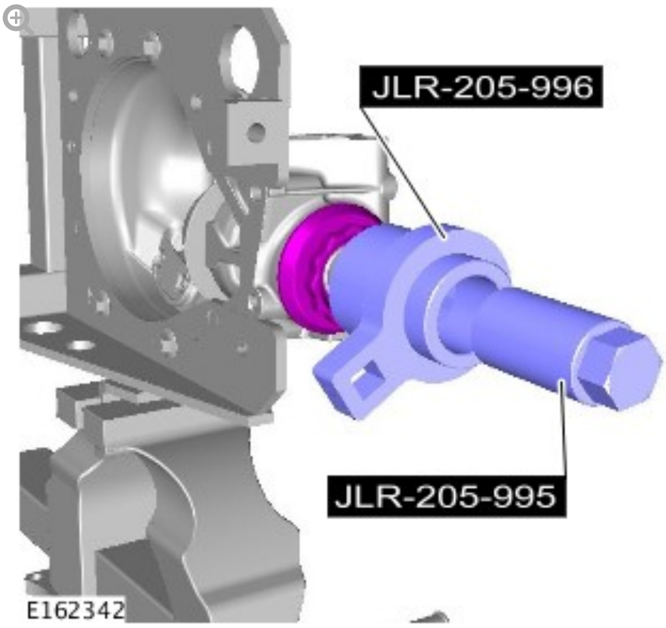


E162341

Install the Special Tool(s): JLR-205-995, 205-067

10. **CAUTION:**

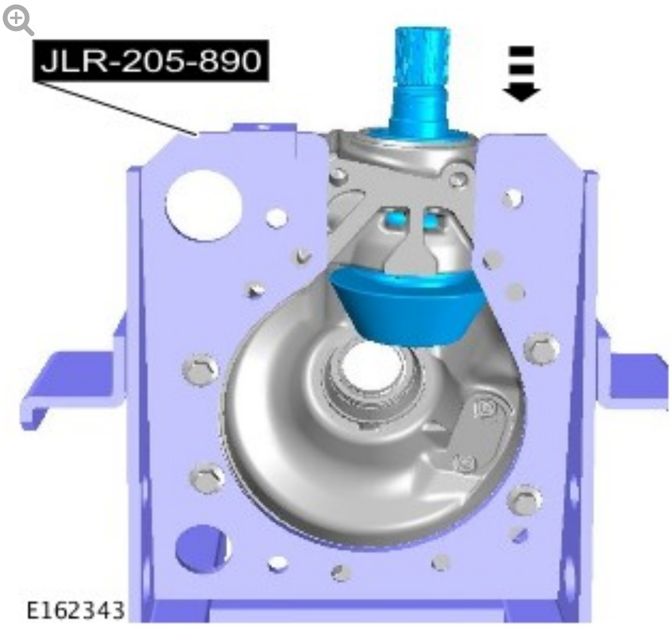
The special tool JLR-205-995 should be rotated clockwise to remove the drive pinion nut.



Install the Special Tool(s): JLR-205-996, JLR-205-995

11. **CAUTION:**

Damage to the casing may occur if the base plates to support the special tool on the press are not used.



Special Tool(s): JLR-205-890

12. **WARNINGS:**

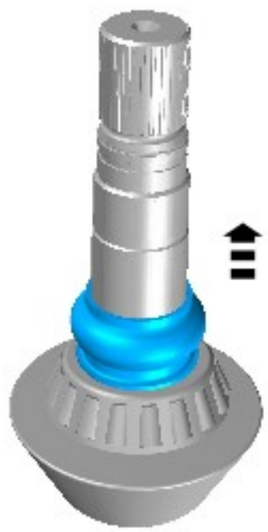
- Do not handle the head bearing.
- Do not clean or degrease the head bearing.

CAUTION:

Check the drive pinion assembly and bearing for any damage.

NOTES:

- Make sure to clean the threads prior to installation using Loctite cleaner 7200.
- Discard the drive pinion spacer.


E162344

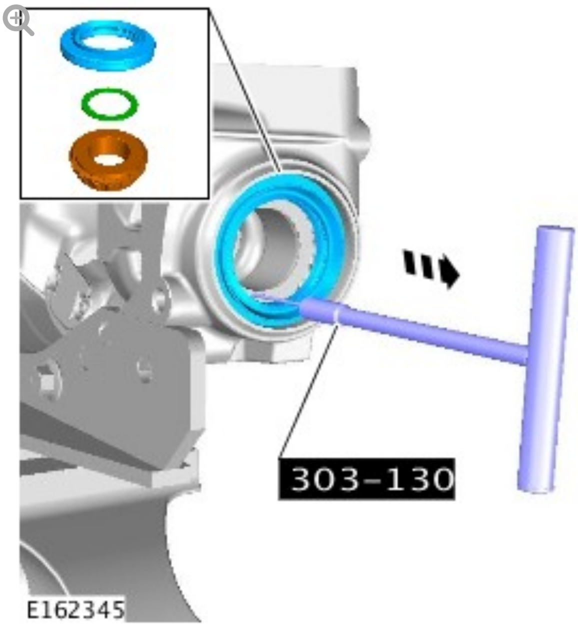
13.

CAUTION:

Make sure the drive pinion seal recess is not damaged as the seal is removed.

NOTE:

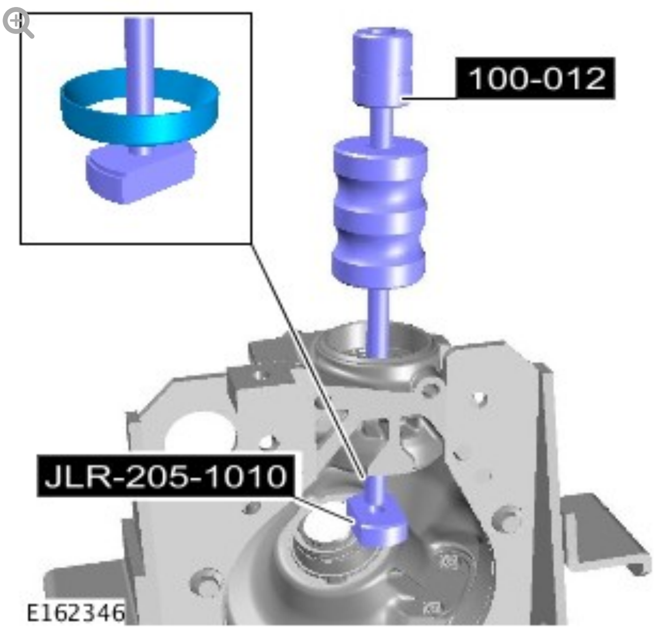
Discard the drive pinion seal, tail bearing thrust washer and tail bearing.



Install the Special Tool(s): 303-1300

14. **WARNING:**

Check special tool for any damage before removing the drive pinion tail bearing outer race.



Using the special tools remove the pinion drive tail bearing outer race.

Special Tool(s): 100-012, JLR-205-1010

15. **CAUTION:**

It is essential that absolute cleanliness is observed when working on the front differential. Always cover any open orifices using lint free non-flocking material to prevent the ingress of foreign matter. Failure to follow this instruction may result damage to the components.

NOTE:

Inspect the drive pinion tail bearing outer race carrier for damage.

Using Loctite cleaner 7200 thoroughly clean the differential casings.

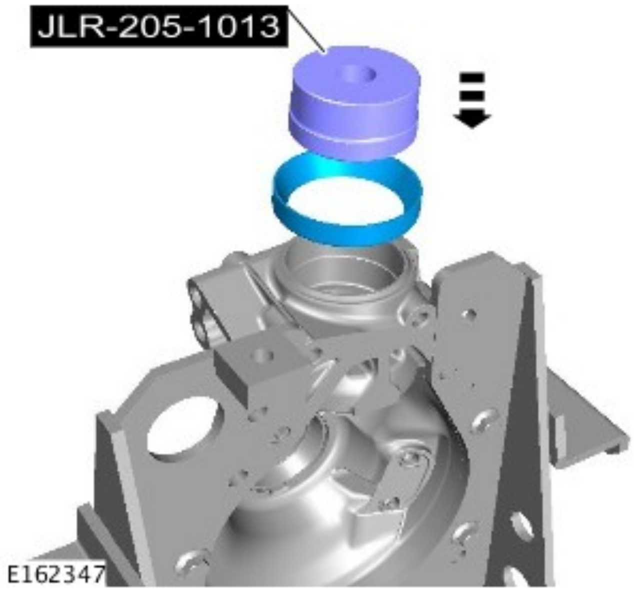
INSTALLATION

1. **WARNING:**

Check the special tool for damage before installing the new drive pinion tail bearing outer race.

CAUTION:

Do not clean or lubricate the new drive pinion tail bearing, as it is supplied coated with a low friction oil. Failure to follow this instruction will require a new drive pinion tail bearing to be installed before the differential can be successfully assembled.

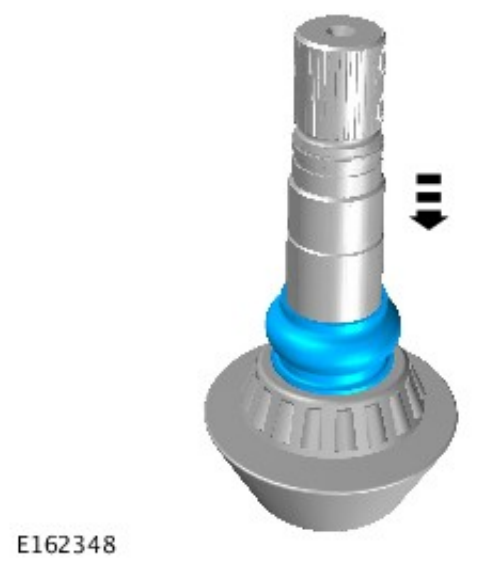


Using a suitable press, install the new drive pinion tail bearing outer race.

Special Tool(s): JLR-205-1013

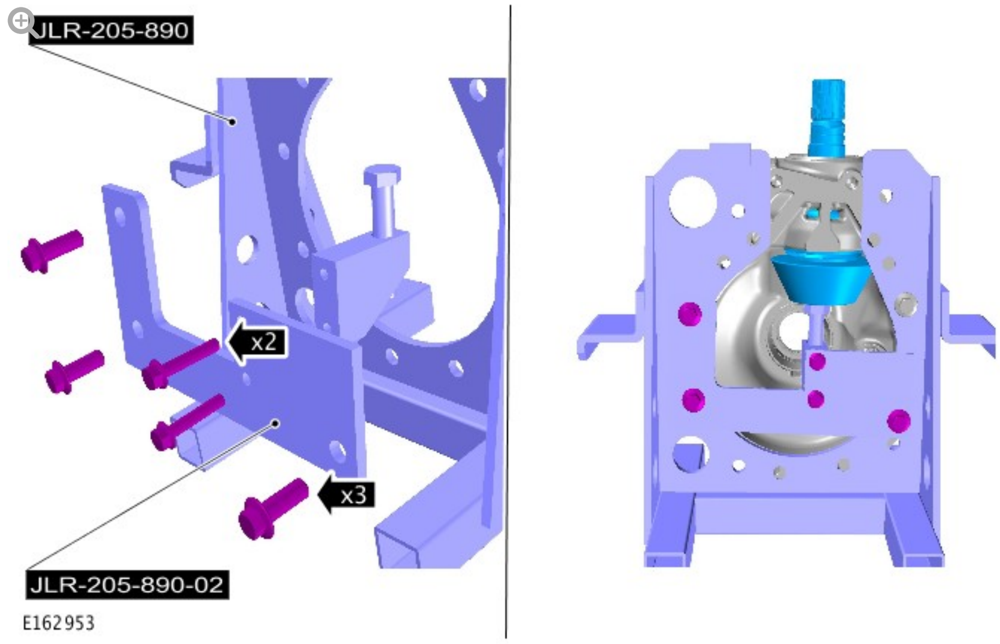
2. **NOTE:**

Make sure to clean the threads prior to installation using Loctite cleaner 7200.



Install the new drive pinion spacer.

3.



Special Tool(s): JLR-205-890, JLR-205-890-2

Torque: **45 Nm**

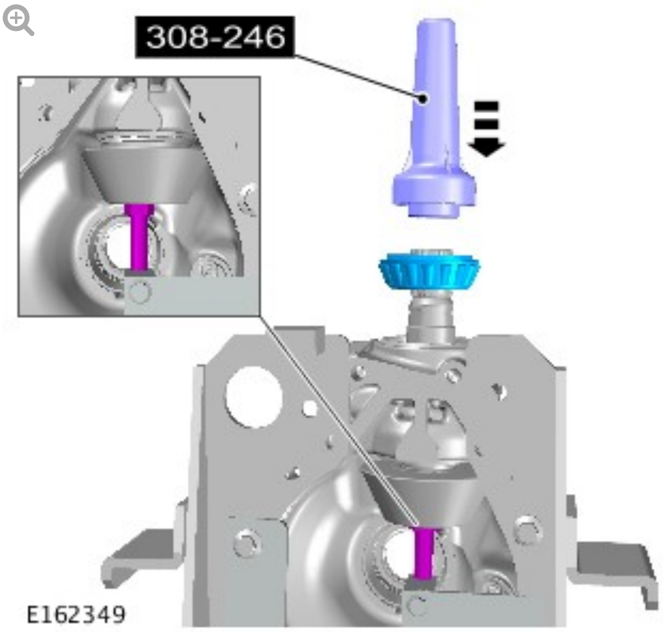
4.

WARNING:

Using the special tool, secure the drive pinion assembly.

CAUTION:

Check the drive pinion assembly is secure before installing the drive pinion tail bearing.



Using a suitable press, install the new drive pinion tail bearing.

Install the Special Tool(s): 308-246

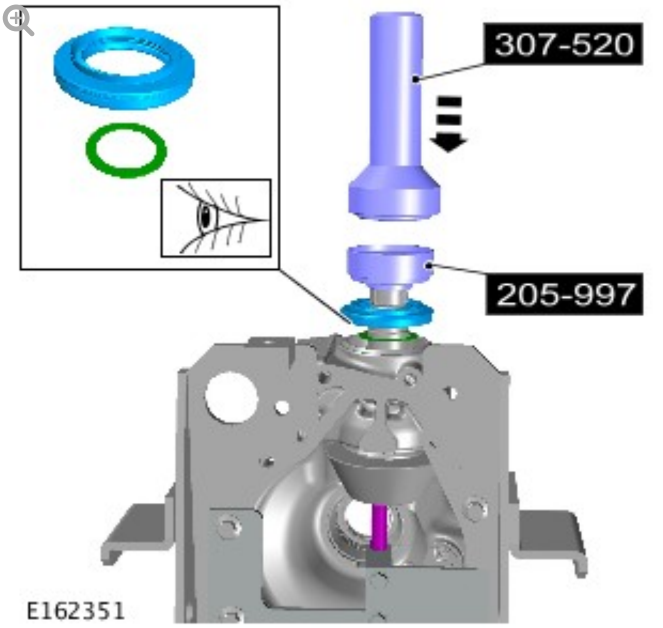
5.

CAUTIONS:

- Check the special tool for damage before installing the new seal.
- Check the special tool is clean and free from foreign material.
- Do not lubricate the drive pinion oil seal.

NOTES:

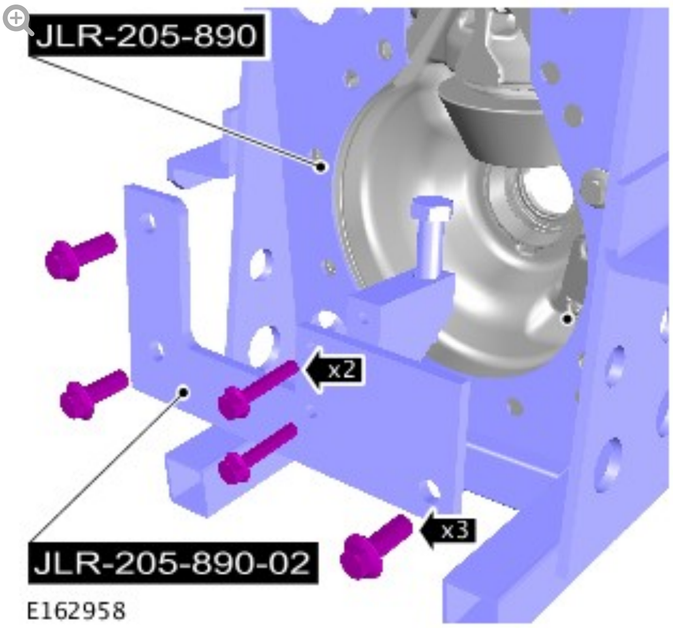
- Using the special tool, secure the drive pinion assembly.
- Install the new drive pinion tail bearing thrust washer.



Using a suitable press, install the new drive pinion seal.

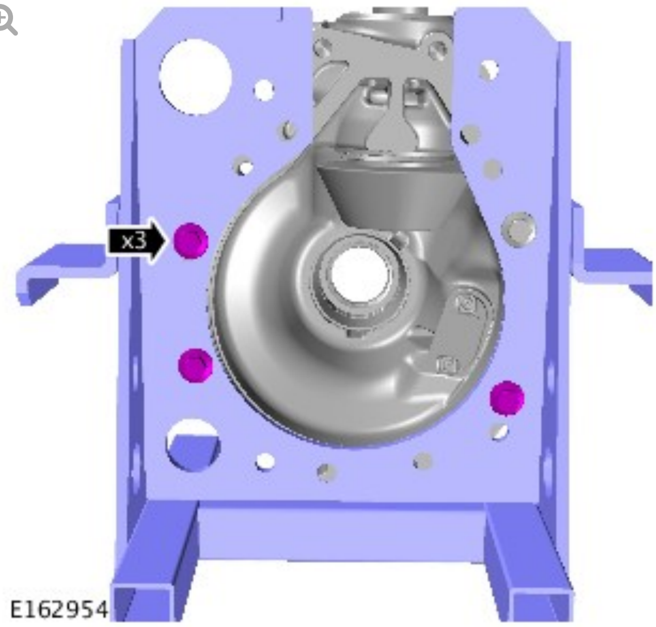
Install the Special Tool(s): JLR-205-997, 307-520

6.



Special Tool(s): JLR-205-890, JLR-205-890-2

7.



Torque: **45 Nm**

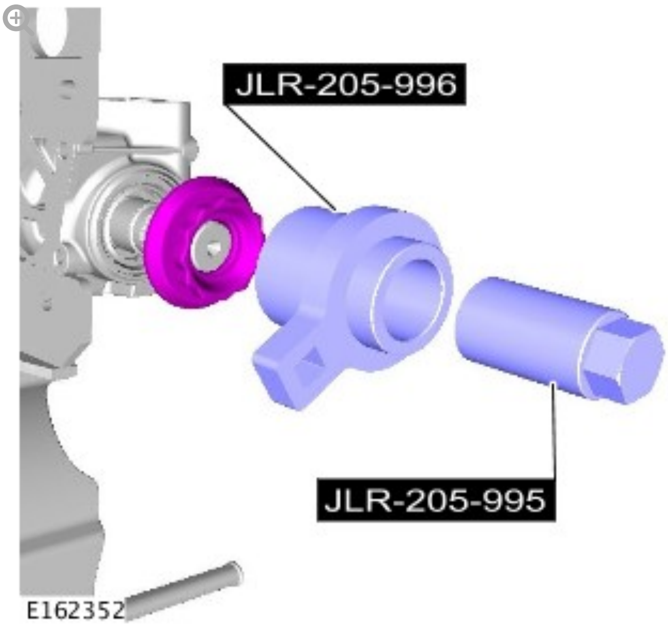
8.

WARNING:

High torque application.

CAUTIONS:

- The special tool JLR-205-995 should be rotated counter clockwise to install the new drive pinion shaft nut.
- Do not use air tools to install the nut. Failure to follow this instruction may result in damage to the component.

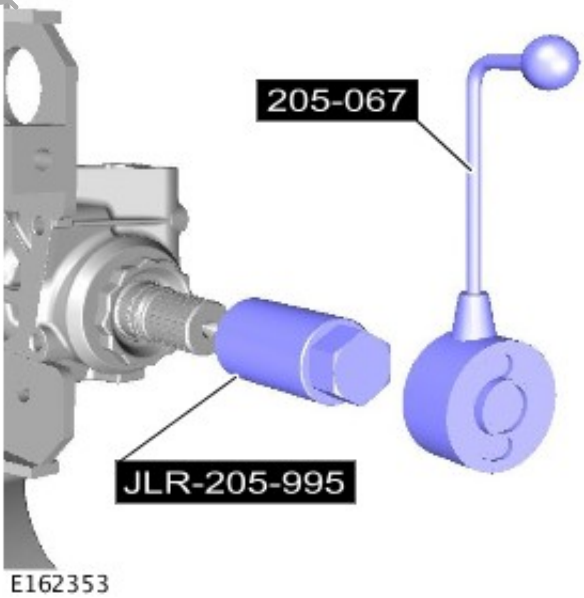


- Apply Loctite 243 to the drive pinion shaft threads.
- Install the new drive pinion nut, tighten the nut until the collapsible spacer begins to collapse.

  Install the Special Tool(s): JLR-205-996, JLR-205-995

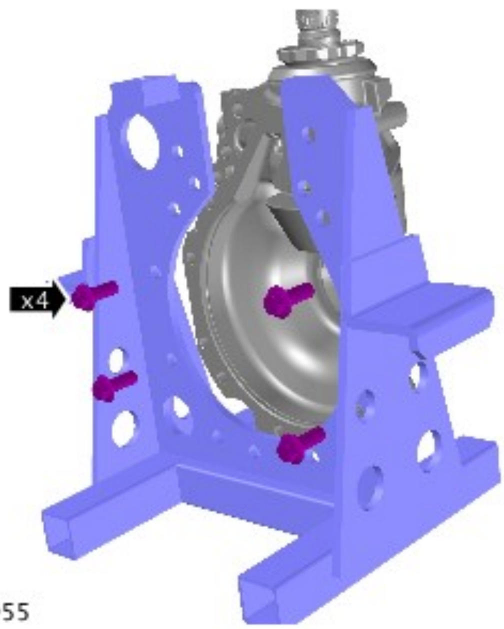
9. **CAUTIONS:**

- Make sure the specified torque to rotate the drive pinion shaft is not exceeded. If excess preload is applied to the joint the drive pinion shaft should be removed and a new collapsible spacer, drive pinion tail bearing, bearing cup and flange nut must be installed.
- The drive pinion shaft should be rotated through two full revolutions by hand before the torque measurement is performed.
- The special tool should be rotated at a constant steady speed to produce a consistent reading.
- The special tool 205-067 should be turned clockwise to check the torque to rotate.



- Check the torque required to rotate the drive pinion nut. The specified torque is the torque to rotate figure recorded in step 9 of the removal process + 0.45 Nm
- If the specified torque to rotate the pinion is not reached, tighten the drive pinion flange nut in one degree increments and check the rotational torque after each until the specified torque is achieved.
  Install the Special Tool(s): 205-067, JLR-205-995

10.



11.

Install the crown wheel assembly.

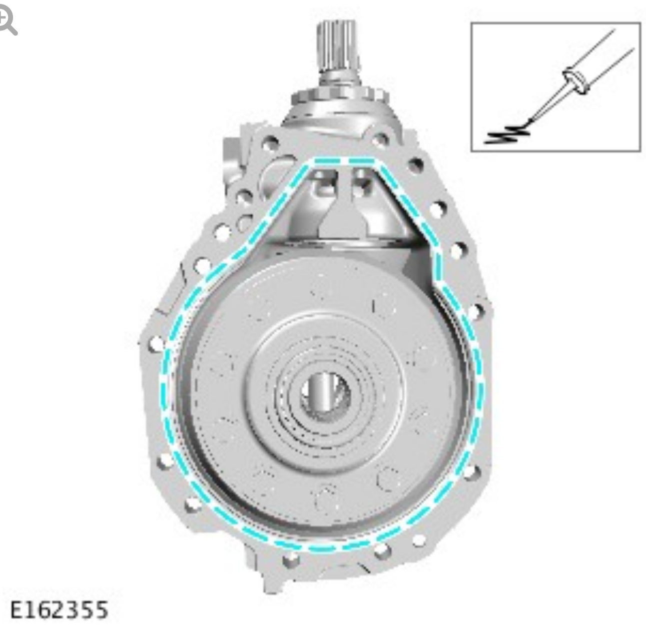
12.

WARNING:

Clean the bolt holes to remove any sealant and make sure the bolts threads are clean.

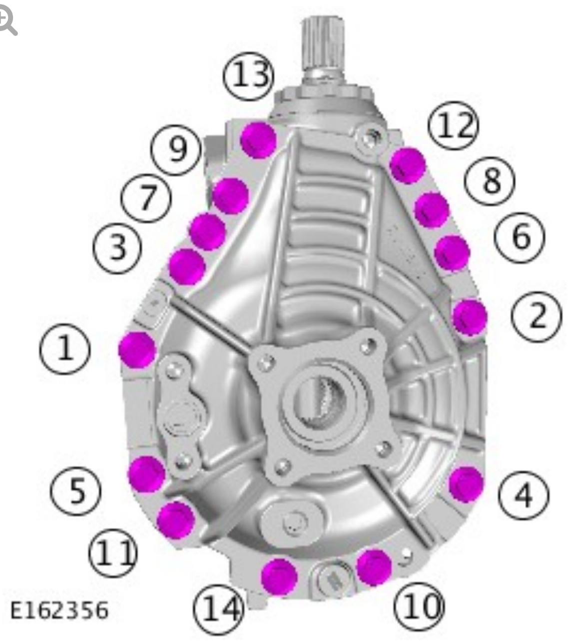
CAUTION:

Using Loctite cleaner 7200 thoroughly clean the mating faces, before the sealant is applied.



Apply a 3 Mm bead of sealant STC50550 to the areas shown.

13.



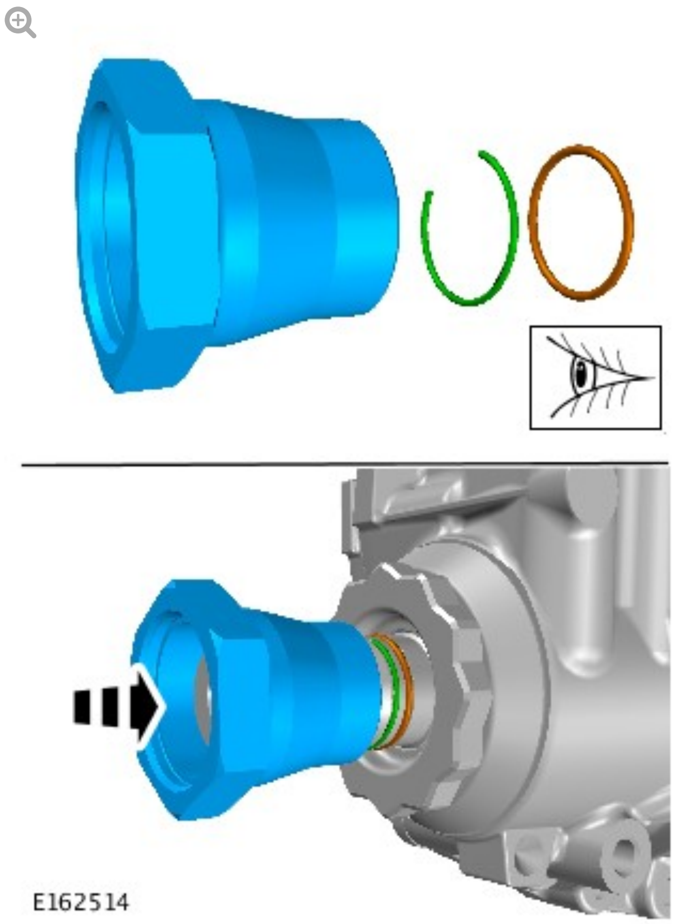
Install the differential casing bolts in the sequence shown.

Torque: **90-95 Nm**

14.

NOTES:

- Install the drive pinion o-ring seals.
- Install the drive pinion circlip.



Install the drive shaft nut.

15. Refer to: Axle Assembly (205-03 Front Drive Axle/Differential, Removal and Installation).

16. Refer to: Differential Draining and Filling (205-03 Front Drive Axle/Differential, General Procedures).